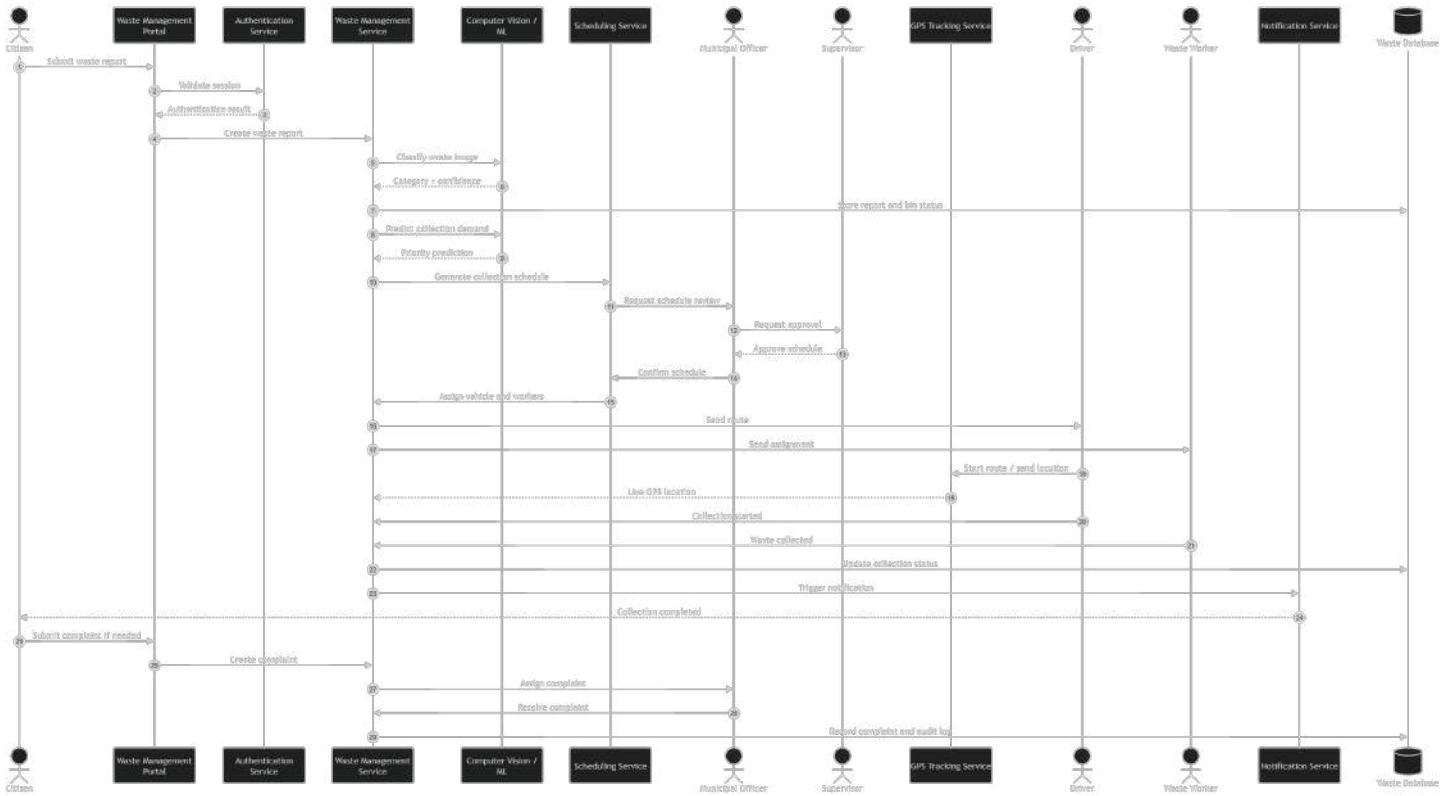
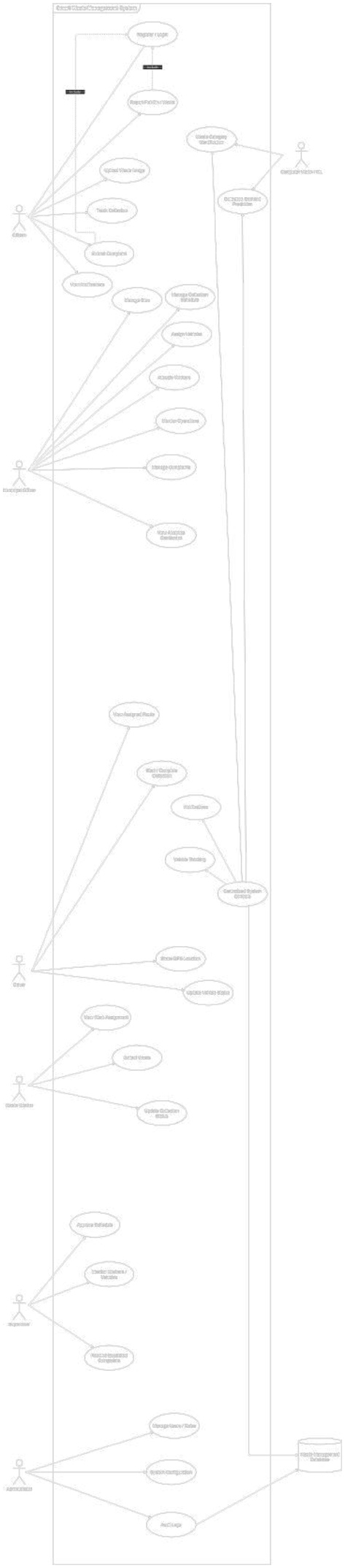
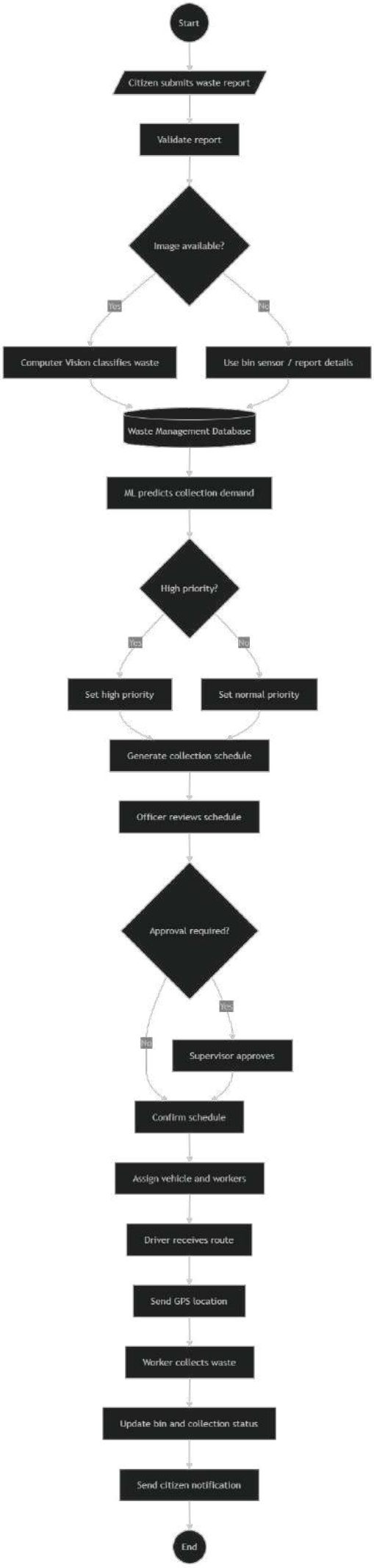
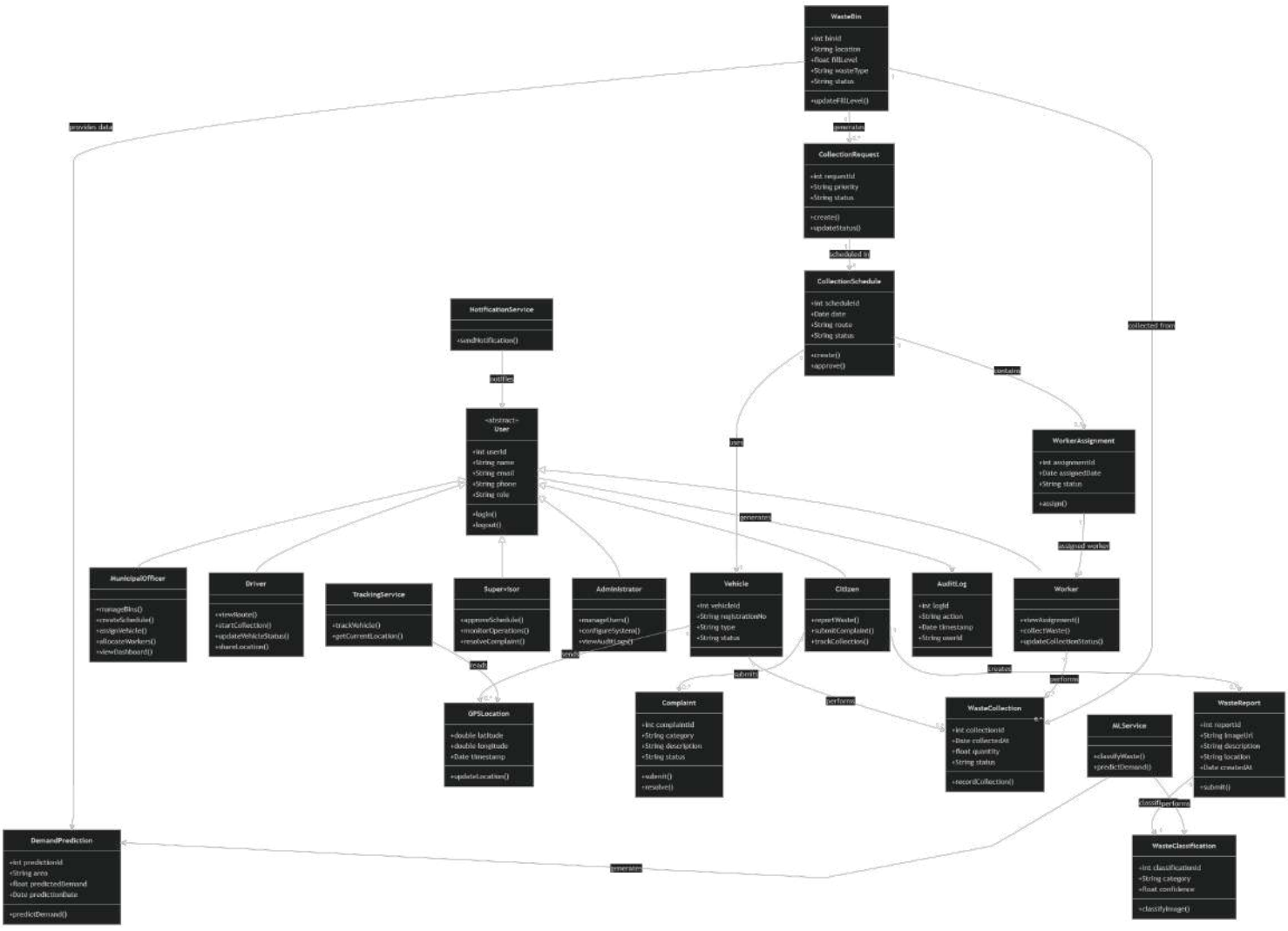
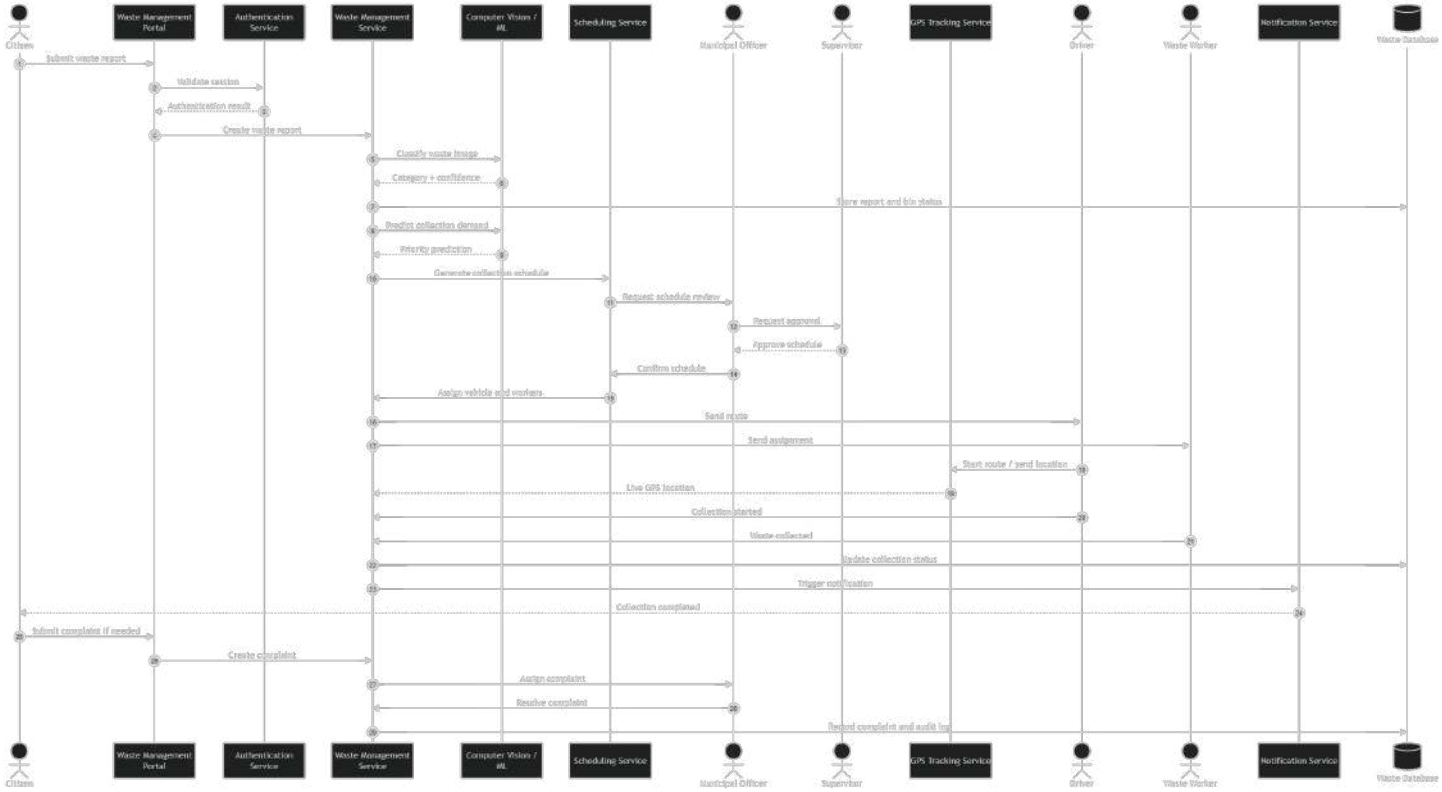

Citizen
Waste Management Portal
Authentication Service
Waste Management Service
Computer Vision / ML
Scheduling Service
Municipal Officer
Supervisor
GPS Tracking Service
Driver
Waste Worker
Notification Service
Waste Database
1 Submit waste report
2 Validate session
3 Authentication result
4 Create waste report
5 Classify waste image
6 Category + confidence
7 Store report and bin status
8 Predict collection demand
9 Priority prediction
10 Generate collection schedule
11 Request schedule review
12 Request approval
13 Approve schedule
14 Confirm schedule
15 Assign vehicle and workers
16 Send route
17 Send assignment
18 Start route / send location
19 Live GPS location
20 Collection started
21 Waste collected
22 Update collection status
23 Trigger notification
24 Collection completed
25 Submit complaint if needed
26 Create complaint
27 Assign complaint
28 Resolve complaint
29 Record complaint and audit log


Smart Waste Management System
Register / Login
Report Full Bin / Waste
Upload Waste Image
Track Collection
Submit Complaint
View Notifications
Citizen
Waste Category Identification
Classification Model Prediction
Manage Bins
Manage Collection Schedule
Assign Vehicles
Activate Workers
Monitor Operations
Manage Complaints
View Analytics Dashboard
Municipal Officer
View Assigned Route
Start / Complete Collection
Mark Bin Empty
Validate Recycling
Generate System Alerts
Share GPS Location
Update Vehicle Status
Driver
View Task Assignment
Scan Bin/ Waste
Update Collection Status
Waste Worker
Approve Schedule
Monitor Citizens / Vehicles
Inspector
Manage Users / Roles
System Configuration
Audit Logs
Administrator
Waste Management Database


Start
Citizen submits waste report
Validate report
Image available?
Yes
No
Computer Vision classifies waste
Use bin sensor / report details
Waste Management Database
ML predicts collection demand
High priority?
Yes
No
Set high priority
Set normal priority
Generate collection schedule
Officer reviews schedule
Approval required?
Yes
No
Supervisor approves
Confirm schedule
Assign vehicle and workers
Driver receives route
Send GPS location
Worker collects waste
Update bin and collection status
Send citizen notification
End


WasteBin
+int binId
+String location
+float fillLevel
+String wasteType
+String status
+updateFillLevel()
provides data
generates
CollectionRequest
+int requestId
+String priority
+String status
+create()
+updateStatus()
scheduled in
CollectionSchedule
+int scheduleId
+Date date
+String route
+String status
+create()
+approve()
collected from
contains
NotificationService
+sendNotification()
notifies
<abstract>
User
+int userId
+String name
+String email
+String phone
+String role
+login()
+logout()
WorkerAssignment
+int assignmentId
+Date assignedDate
+String status
+assign()
assigned worker
MunicipalOfficer
+manageBins()
+createSchedule()
+assignVehicles()
+allocateWorkers()
+viewDashboard()
Driver
+viewRoute()
+startCollection()
+updateVehicleStatus()
+shareLocation()
TrackingService
+trackVehicle()
+getCurrentLocation()
Supervisor
+approveSchedule()
+monitorOperations()
+resolveComplaint()
Administrator
+manageUsers()
+configureSystem()
+viewAuditLogs()
Vehicle
+int vehicleId
+String registrationNo
+String type
+String status
Citizen
+reportWaste()
+submitComplaint()
+trackCollection()
AuditLog
+int logId
+String action
+Date timestamp
+String userId
Worker
+viewAssignment()
+collectWaste()
+updateCollectionStatus()
assigns
generates
submits
performs
created
performs
reads
GPSLocation
+double latitude
+double longitude
+Date timestamp
+updateLocation()
Complaint
+int complaintId
+String category
+String description
+String status
+submit()
+resolve()
WasteCollection
+int collectionId
+Date collectedAt
+float quantity
+String status
+recordCollection()
MLService
+classifyWaste()
+predictDemand()
WasteReport
+int reportId
+String imageUrl
+String description
+String location
+Date createdAt
+submit()
classifies/forms
DemandPrediction
+int predictionId
+String area
+float predictedDemand
+Date predictionDate
+predictDemand()
generates
WasteClassification
+int classificationId
+String category
+float confidence
+classifyImage()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